Ali Hassan

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Professional Summary

Embedded Systems Engineer with experience in firmware development, robotics, PCB design, and IoT applications. Skilled in C/C++, FreeRTOS, Zephyr, STM32, ESP32, and embedded communication protocols. Experienced in sensor integration, hardware–software co-design, and real-time system validation.

Experience

Robotics & Embedded Systems Engineer Intern, ABMI Groupe — Lyon, France — Mar 2026 – Present

- Developed 7 ROS 2 communication nodes for real-time sensor integration, system control, and autonomous coupling workflows, achieving 55 ms end-to-end latency.
- Designed PID-based control algorithms and integrated UWB, IMU, and vision sensors for vehicle–trailer alignment, achieving 10 cm localization accuracy.
- Built a GUI monitoring tool and supported experimental validation of a 6-DOF Stewart platform.

Embedded Systems Engineer, MindTune Innovations — Wah Cantt, PK — Mar 2025 – Sep 2025

- Developed IoT firmware on Zephyr RTOS and FreeRTOS for wireless EEG earbuds, reducing power draw by 15%.
- Built a wireless EEG acquisition system (ADS1299 + nRF52840), streaming 3-channel data over BLE at 90 ms latency.
- Optimized firmware for wireless earbuds on a Jieli AC6966 SoC, reducing audio latency to 40 ms and improving battery life by 10%.

Embedded Systems Engineer, Revive Medical Technologies — Islamabad, PK — Aug 2024 – Mar 2025

- Developed and unit-tested firmware across PIC, STM32, and nRF microcontrollers for medical devices, achieving 100% test coverage and zero critical defects.
- Implemented UART, SPI, I2C, and IoT-based communication between medical sensors and MCU for remote-monitoring deployments.

Technical Skills

- **Programming:** C, C++, Embedded C, Python, MATLAB
- **Embedded Platforms:** STM32 (ARM Cortex-M), ESP32, nRF52, PIC, Raspberry Pi, bare-metal & RTOS
- **RTOS & Firmware:** FreeRTOS, Zephyr RTOS, Device Drivers, Firmware Development
- **Communication Protocols:** UART, SPI, I2C, CAN, USB, RS-485, BLE, Wi-Fi, Modbus RTU, Zigbee, LoRaWAN
- **Robotics & Control:** ROS 2, PID Control, Sensor Fusion, Localization, Motor Control, Encoder Feedback
- **Hardware & Validation:** PCB Design, Schematic Design, Sensor Integration, Testing, Debugging
- **Tools:** Altium Designer, MATLAB/Simulink, Linux, Git, GitHub, GitHub Actions

Education

Master's in Embedded Systems Security *(Bac+5)* — *Sep 2025 – Sep 2026*
Grenoble INP Esisar, Valence, France

BS Electrical Engineering — *Sep 2020 – Aug 2024*
COMSATS University Islamabad, Pakistan

Projects

ISE to Modbus RTU Converter — GitHub

- Designed an industrial Modbus RTU converter with isolated power supply, 24-bit ADC sensing, and RS-485 communication, supporting 3 sensor channels.

Catheter Trackability Testing Machine — GitHub

- Developed an embedded medical testing system for catheter force measurement using load-cell sensing and motor control, achieving 20 N force-measurement resolution.

EEG-Integrated Wireless Earbuds — GitHub

- Developed EEG-integrated TWS wireless earbuds combining AC6966B Bluetooth audio firmware, ADS1299-based 6-channel biosignal acquisition, and nRF52 BLE telemetry for synchronized 250 SPS mobile-app streaming.

Autonomous Coupling System Using a 6-DOF Stewart Platform — GitHub

- Built a ROS 2-based Stewart platform integrating ArUco vision, IMU sensor fusion, motor control, and encoder feedback, achieving 10 cm coupling alignment accuracy.

5-DoF Robotic Arm System for Skin Tumor Detection and Intervention — GitHub

- Built a Raspberry Pi–based system integrating a CNN and OpenCV to detect skin tumors with 87% accuracy, extract target coordinates, and guide a 5-DoF robotic arm for controlled intervention. .

Languages

English: C1 | **French:** A1